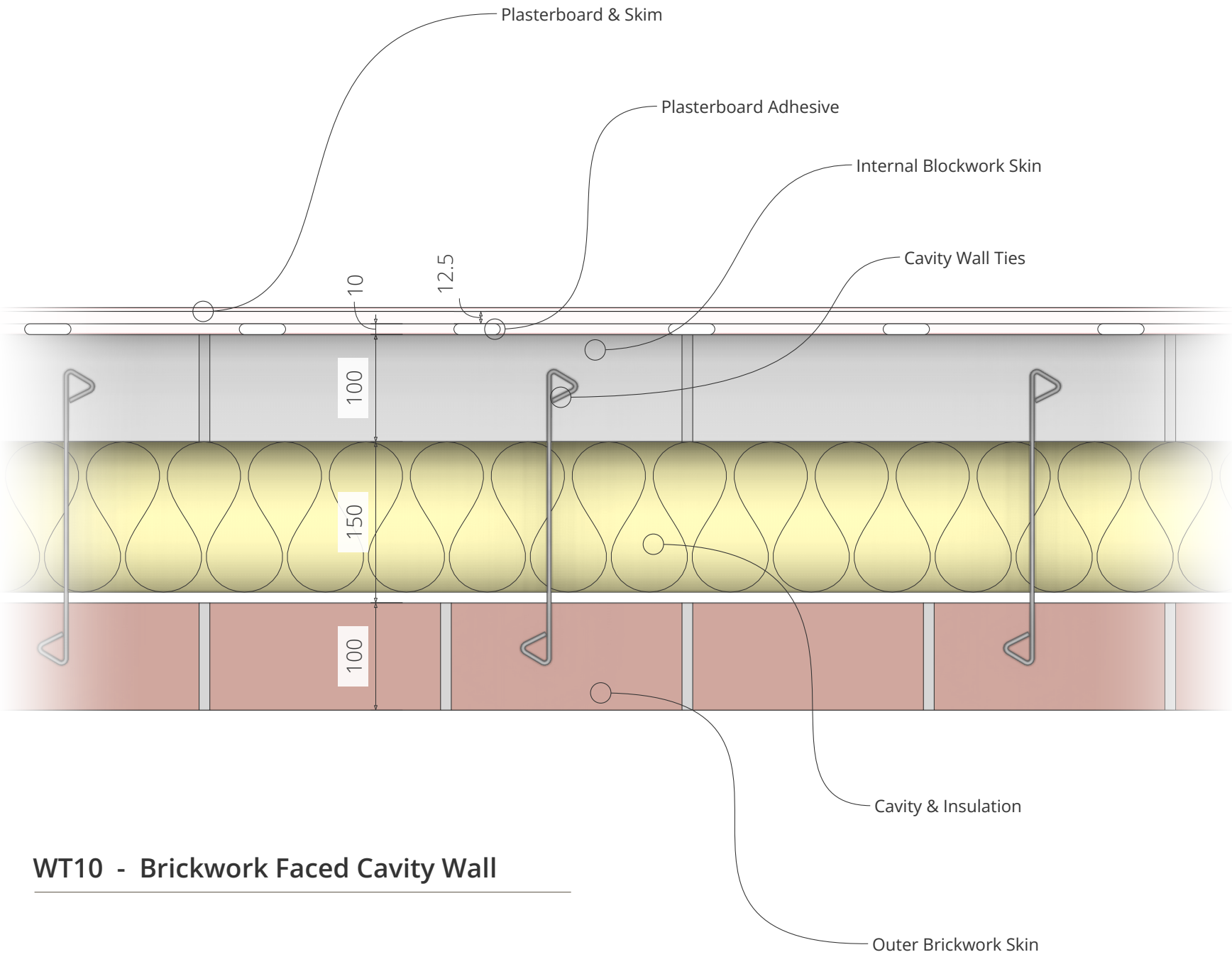




WT10 - Brickwork Faced Cavity Wall

| Calculation                                                                             |                |               |                 |
|-----------------------------------------------------------------------------------------|----------------|---------------|-----------------|
| U = 1 / (0.13 + 0.12 + 6.36 + 0.06 + 0.02 + 0.05 + 0.01 + 0.04) = 1 / 6.80 = 0.15 W/m²K |                |               |                 |
|                                                                                         |                |               |                 |
| Layer                                                                                   | Thickness (mm) | Lambda (W/mK) | R-value (m²K/W) |
| Internal Surface Resistance (Rsi)                                                       | -              | -             | 0.130           |
| Facing Brickwork Wall Skin                                                              | 102.5          | 0.85          | 0.121           |
| PIR Insulation - Recticel Eurowall Plus Full Fill                                       | 140            | 0.022         | 6.364           |
| Dense Concrete Block Wall Skin                                                          | 100            | 1.56          | 0.064           |
| Plasterboard Adhesive                                                                   | 10             | 0.46          | 0.022           |
| Plasterboard                                                                            | 12.5           | 0.24          | 0.052           |
| Plaster Skim Coat                                                                       | 3              | 0.46          | 0.006           |
| External Surface Resistance (Rse)                                                       | -              | -             | 0.040           |
| Total R-value                                                                           |                |               | 6.799           |
|                                                                                         |                |               |                 |
| Result                                                                                  |                |               |                 |
| U-Value                                                                                 |                |               | 0.15 W/m²K      |

Value Compliant - England Part L1B : Wall U-value 0.18 W/m²K.

| Element                 | Specifications                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Outer Brickwork Skin    | Facing brick outer leaf to match the host dwelling in colour, texture, and size, with bond, coursing, joint thickness, and pointing profile to visually align where the new work is seen. Mortar to BS EN 998-2 and workmanship to PD 6697 and the principles of BS EN 1996 (Eurocode 6). Set out from agreed datum lines and blend brick packs to avoid banding. Keep the cavity face clean as work proceeds, with no surplus mortar protruding into the cavity. Protect fresh brickwork from rain, frost, and rapid drying. Do not proceed beyond a sample panel (brick and mortar) approved as matching the host.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Cavity & Insulation     | 150mm cavity comprising 140mm Recticel Eurowall PIR partial-fill cavity boards with tongue-and-groove edges and drainage/capillary grooves, installed to maintain a clear 10mm residual cavity to the outer leaf. Insulation boards to be from the current certified range for the intended use (BBA or equivalent third-party certification current at time of installation) and installed strictly to the manufacturer's instructions. Boards to be tightly butted and fully interlocked with joints closed, vertical joints staggered, and continuity maintained at corners and around openings with proprietary closers as detailed. Keep boards clean, dry, and undamaged; replace any bowed, or damaged boards. Maintain the full residual cavity clear of mortar snots, debris, and obstructions to preserve moisture performance and prevent bridging.                                                                                                                                                                                                                                                                         |
| Cavity Wall Ties        | Ancon stainless steel wall ties or approved equivalent to BS EN 845-1, specified as Type 2 general purpose ties suitable for the exposure and build-up. Tie length to suit 150mm cavity, typically 275mm to achieve minimum 62.5mm embedment into each leaf (confirm exact length to manufacturer tables). Stainless steel to be austenitic Grade 304 (EN 1.4301 / A2) minimum. Install ties at 900mm horizontal centres and 450mm vertical staggered centres, with additional ties at unbonded reveals and around openings at 225mm vertical centres, and position ties within 150mm of reveals/edges in accordance with best practice and manufacturer guidance. Fit insulation retaining clips/discs to all ties to mechanically secure the PIR boards tight to the inner leaf without compressing the insulation, while maintaining the specified 10mm residual cavity to the outer leaf. Ties to be kept clear of mortar droppings and laid with a slight fall toward the outer leaf to shed water. Do not use galvanized ties, and do not bend ties to compensate for poor set-out as this can reduce embedment and performance. |
| Internal Blockwork Skin | 100mm dense concrete block inner leaf to BS EN 771-3, nominal 440mm x 215mm units, minimum compressive strength 7.3 N/mm² (commonly referenced as 7N) and approximately 1850 kg/m³ density, laid in mortar to BS EN 998-2. Construct plumb, level, and true with fully filled bed and perpendicular joints to improve airtightness and provide a consistent background for finishes. Strike/point joints neatly on the internal face where required to reduce air leakage paths behind plasterboard and to avoid voids that compromise dab adhesion. Coordinate any structural restraints, lintels, and movement detailing as shown on drawings and engineer's information.                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Interior Wall Finishing | The Contractor is to line general masonry wall areas with 12.5mm standard gypsum plasterboard. However, where linings are situated in close proximity to or encasing structural steel members, the specification must be upgraded to 12.5mm Knauf Fire Panel (or equivalent Type F board compliant with BS EN 520) to ensure the required fire resistance integrity is maintained. All boards are to be fixed on gypsum-based adhesive dabs to a clean, sound, and dry background, with continuous perimeter dabs applied to strictly limit air movement behind the boards in accordance with Approved Document L. The installation is to be finished with a 3mm skim coat of Thistle Multi-Finish plaster (or equivalent to BS EN 13279-1) to achieve a smooth, paint-ready surface.                                                                                                                                                                                                                                                                                                                                                  |
| Design Rationale        | The wall build-up is intended to exceed minimum elemental thermal expectations for extension work to help offset elevated heat loss associated with the high glazing ratio at the gabled elevation, improving comfort and reducing overall fabric heat loss. A high-performance PIR partial-fill cavity board (Recticel Eurowall) is specified to maximise thermal resistance within the available wall thickness while retaining a controlled residual cavity for moisture management and robustness. Dense 7.3 N/mm² blocks are specified for improved durability, acoustic mass, and fixability for domestic use; any reduction in block thermal performance compared with lightweight blocks is more than offset by the increased thermal resistance of the upgraded 140mm cavity insulation. Any change to insulation type, thickness, cavity strategy, or tie specification must be checked for U-value, moisture/condensation risk, and system compatibility before implementation.                                                                                                                                             |

Building Control Approval Submission

This drawing is submitted for Building Control approval. Construction must not commence until the Inspector sanctions the proposed structural details, thermal performance, and moisture integrity. The Principal Contractor is fully responsible for coordinating all statutory inspections to verify strict compliance with the Building Regulations 2010.